

MINSEOK JOO

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RESEARCH INTERESTS

Generative AI for modeling the structure and dynamics of the real world, with a focus on **video world models**, **video generation**, and **3D vision**.

EDUCATION

Korea University Seoul, Republic of Korea
Integrated M.S.–Ph.D. Program in Computer Science and Engineering Feb 2022–Present
Advisor: [Prof. Hyunwoo J. Kim](#) (KAIST)

Korea University Seoul, Republic of Korea
B.S. in Biomedical Engineering Mar 2016–Feb 2022
Second Major: Artificial Intelligence

RESEARCH EXPERIENCE

Machine Learning & Vision Lab Feb 2022–Present
Korea Advanced Institute of Science and Technology (KAIST)
Graduate Researcher | Advisor: [Prof. Hyunwoo J. Kim](#)

- Research on retrieval-augmented long-video generation and geometric consistency (COVRAG; NeurIPS 2026, accepted).
- Joint work on robust camera–LiDAR 3D object detection (MEFormer; co-first author), with additional research on 3D representations and generative models.

BREIN Lab, Korea University Jun 2020–Jun 2021
Undergraduate Researcher | Advisor: Prof. Joon-Kyung Seong

- Investigated graph neural networks for decomposing resting-state fMRI (rs-fMRI) brain networks and analyzing salient functional connectivity.

RESEARCH PROJECTS

3D High-level Scene Understanding and Situation Prediction for Urban Drones 2023–Present
Participating Researcher | NRF Mid-Career Research Program

- Collect and preprocess urban drone data from GTA simulations; investigate 3D scene understanding, relationship prediction, and motion prediction.
- Research agent-based automatic labeling for urban drone data.

Funding: National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT).

Computer Vision for Breast Cancer Biomarker Development 2023
Research Collaborator | Korea University Ansan Hospital

- Applied computer vision to quantify extracellular matrix components and detect and count immune cells in fluorescence microscopy images, supporting breast cancer biomarker development.

HONORS & AWARDS

Minister's Award, 3rd Place 2021
AI Grand Challenge for Disaster Response Drones, Ministry of Science and ICT, Republic of Korea

TEACHING EXPERIENCE

Teaching Assistant, Korea University 2023–2024
Artificial Intelligence Spring 2023, Spring 2024
Deep Learning Fall 2023, Fall 2024

Teaching Assistant, LG AI Workshop Jan 2023

ACADEMIC SERVICE

Conference Reviewer: ICML 2026; ECCV 2026; NeurIPS 2026; ICLR 2027.

*Equal contribution.

Conference Papers

- [C5] **Retrieve What’s Missing: Coverage-Maximizing Retrieval for Consistent Long Video Generation**
Minseok Joo, Dogyun Park, Taehoon Lee, Kyujin Lee, Hyunwoo J. Kim
Advances in Neural Information Processing Systems (NeurIPS), 2026. Accepted. [Paper]
- [C4] **F⁴Splat: Feed-Forward Predictive Densification for Feed-Forward 3D Gaussian Splatting**
Injae Kim, Chaehyeon Kim, Minseong Bae, **Minseok Joo**, Hyunwoo J. Kim
European Conference on Computer Vision (ECCV), 2026. [Paper] [Project] [Code]
- [C3] **Blockwise Flow Matching: Improving Flow Matching Models For Efficient High-Quality Generation**
Dogyun Park, Taehoon Lee, **Minseok Joo**, Hyunwoo J. Kim
Advances in Neural Information Processing Systems (NeurIPS), 2025. [Paper] [Code]
- [C2] **Generative Subgraph Retrieval for Knowledge Graph–Grounded Dialog Generation**
Jinyoung Park, **Minseok Joo**, Joo-Kyung Kim, Hyunwoo J. Kim
Conference on Empirical Methods in Natural Language Processing (EMNLP), 2024. [Paper] [Code]
- [C1] **Semantic-Aware Implicit Template Learning via Part Deformation Consistency**
Sihyeon Kim, **Minseok Joo**, Jaewon Lee, Juyeon Ko, Juhan Cha, Hyunwoo J. Kim
IEEE/CVF International Conference on Computer Vision (ICCV), 2023. [Paper] [Code]

Journal Articles

- [J1] **Optimized prediction of breast cancer tumor microenvironment using MRI-based intratumoral and peritumoral radiomics: a prospective study**
Eun Sil Kim, Sungwon Ham, Bo Kyoung Seo, Ji Young Lee, Woong Sun, Minkyu Jeon, **Minseok Joo**, Seonghoon Park, Shuncong Wang, Boram Lee, Hye Yoon Lee, Min Sun Bae, Kyu Ran Cho, Ok Hee Woo, Sung Eun Song, Soo-Yeon Kim
Frontiers in Oncology, 15:1654508, 2025. [Paper]
In collaboration with Korea University Ansan Hospital.

Preprints

- [P3] **Retrieval-Augmented Skill Optimization via Cross-Harness Adaptation**
Jaewon Chu, Ji Soo Lee, Jihwan Park, Dohwan Ko, Jeehye Na, Seunghun Lee, Taehoon Lee, Minseo Yoon, **Minseok Joo**, Yunyang Xiong, Hyunwoo J. Kim
arXiv:2609.38024, 2026. [Paper]
- [P2] **Mitigating State Aliasing in Vision-Language-Action Models via Inverse Dynamics Learning**
Kyujin Lee, Injae Kim, Jihwan Park, Yejun Ju, **Minseok Joo**, Hyunwoo J. Kim
arXiv:2605.29577, 2026. [Paper]
- [P1] **Robust Multimodal 3D Object Detection via Modality-Agnostic Decoding and Proximity-based Modality Ensemble**
Juhan Cha*, **Minseok Joo***, Jihwan Park, Sanghyeok Lee, Injae Kim, Hyunwoo J. Kim
arXiv:2407.19156, 2024. [Paper] [Code]